Uber Booking Analytics

Business Problem Statement

• The company cannot clearly track how many rides are successfully completed vs cancelled or incomplete, which affects service reliability.

• It is difficult to understand who is cancelling rides (customer or driver), making it hard to fix the main issues.

• Service quality is not consistent across different vehicle types, which affects customer experience.

• The company is losing money due to cancelled and incomplete rides, but the exact loss is not clearly measured.

• There is limited understanding of how different vehicle types are performing, leading to inefficient use of drivers and cars.

• The business is not fully using data to identify loyal or frequent customers, missing opportunities for retention and growth.

• Some rides remain incomplete without clear reasons, showing gaps in operations that need to be fixed.

Core Objective:
The goal is to use data to improve ride success, reduce cancellations, increase customer satisfaction, and grow revenue.

Project Overview

This project focuses on analyzing Uber booking data to understand ride performance, customer behavior, and overall business efficiency. The goal is to identify key issues such as ride cancellations, incomplete trips, and variations in service quality across different vehicle types.

Using SQL for data analysis and visualization tools like Power BI, the project explores important metrics such as booking success rate, cancellation patterns, customer and driver ratings, and revenue generated from successful rides.

The analysis helps uncover insights into why rides fail, which vehicle types perform better, and how customer engagement impacts business growth. These insights can be used to improve ride reliability, enhance customer experience, optimize fleet usage, and increase overall revenue.

Dataset Summary

• Rows: ~40,000+

• Columns: 20+

• Booking details: Booking ID, Booking Status, Date, Time

• Customer information: Customer ID

• Ride details: Vehicle Type, Pickup Location, Drop Location, Ride Distance

• Time-based features: Hour, Day, Peak Indicator (Is_Peak)

• Cancellation data: Cancelled Rides by Customer, Cancelled Rides by Driver, Cancellation Reasons

• Incomplete ride data: Incomplete Rides, Incomplete Ride Reasons

• Performance metrics: Avg VTAT (Vehicle Time to Arrival), Avg CTAT (Customer Time to Arrival)

• Ratings: Driver Ratings, Customer Rating

• Revenue data: Booking Value

Data Cleaning using Excel

• Cleaned and standardized raw Uber booking data using Microsoft Excel to ensure accuracy and consistency.

• Removed duplicate records and handled missing/null values in key columns like ratings, booking status, and ride details.

• Standardized text fields (e.g., booking status, vehicle type, cancellation reasons) to maintain uniform naming.

• Converted date and time columns into proper formats and created additional fields like hour, day, and peak indicators for analysis.

• Checked and corrected data inconsistencies in numerical fields such as booking value, ride distance, and ratings.

• Ensured all column names were clean, consistent, and analysis-ready (e.g., lowercase or formatted properly for SQL).

• Validated dataset to make it ready for SQL querying and Power BI visualization.

• SQL Integration**:** Loaded the cleaned and transformed Uber booking dataset into a SQL database to perform advanced analysis, such as identifying cancellation patterns, booking success rates, revenue trends, customer behavior, and performance across different vehicle types.

Data Analysis using SQL (Business Questions)

• Retrieve all successful bookings – Identified all the successful booking in dataset

• Average Ride Distance Per Vehicle – Identified Avg Ride Distance for each vehicle type

	Vehicle_Type	avg_distance
▶	Prime SUV	12.914786342689563
	Auto	13.091425425425436
	Mini	13.014270938957626
	Prime Sedan	12.906835043409632
	Prime Plus	13.001066050993765
	eBike	12.96821170809943

	Vehicle_Type	avg_distance
	Auto	13.091425425425436
	Mini	13.014270938957626
	Prime Sedan	12.906835043409632
	Prime Plus	13.001066050993765
	eBike	12.96821170809943
	Bike	12.971665320952763

• Top 5 Customers with Highest Rides – Identified Top 5 Customers who booked the most rides.

	Customer_ID	total_rides
▶	CUST4188	13
	CUST5566	13
	CUST6539	12
	CUST1157	12
	CUST6739	12

• Max & Min Ratings for Prime Sedan Bookings – Identified Max and Min Ratings for bookings of Prime Sedan Category.

	max_rating	min_rating
▶	5	3

• Average Customer Rating As Per Vehicle Type– Identified Average Customer Rating for each Vehicle Type.

	Vehicle_Type	avg_customer_rating
▶	Prime SUV	3.9984080435693166
	Auto	3.9944544544544462
	Mini	4.007564388562152
	Prime Sedan	3.990094711917922
	Prime Plus	4.0044970889379625
	eBike	3.9996190858059375

	Vehicle_Type	avg_customer_rating
	Auto	3.9944544544544462
	Mini	4.007564388562152
	Prime Sedan	3.990094711917922
	Prime Plus	4.0044970889379625
	eBike	3.9996190858059375
	Bike	3.9999192571659226

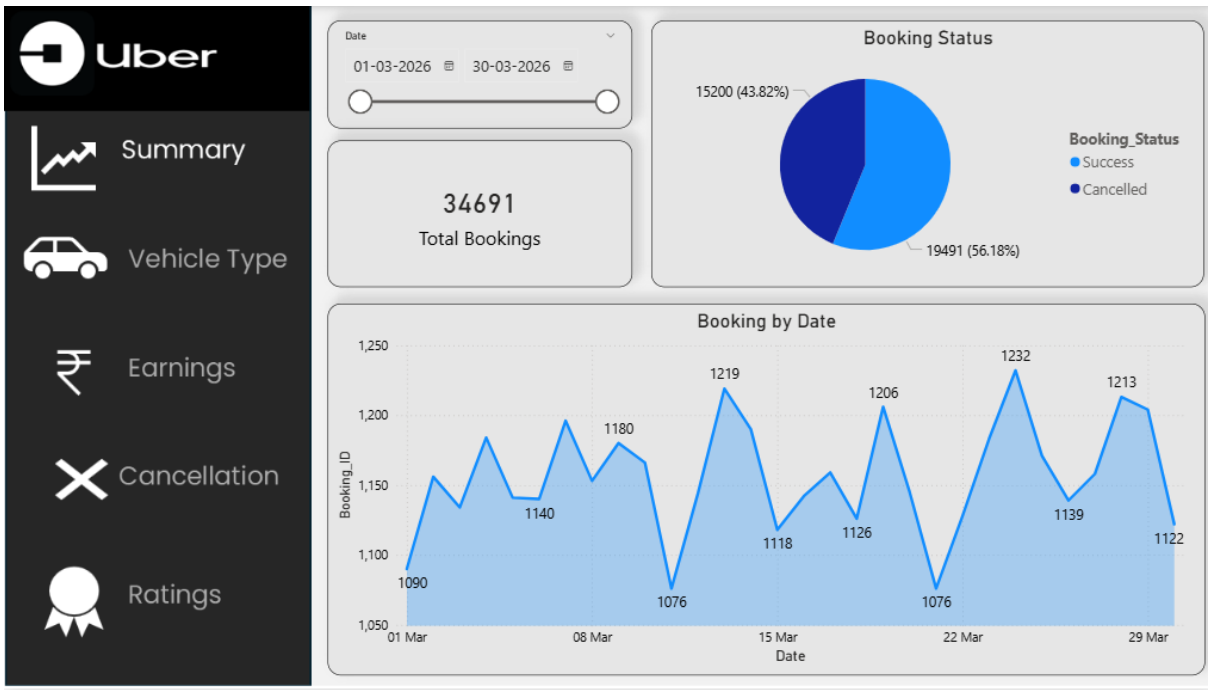
• Total Booking Value for Successful Rides – Identified Total Booking Values for all of the successful rides.

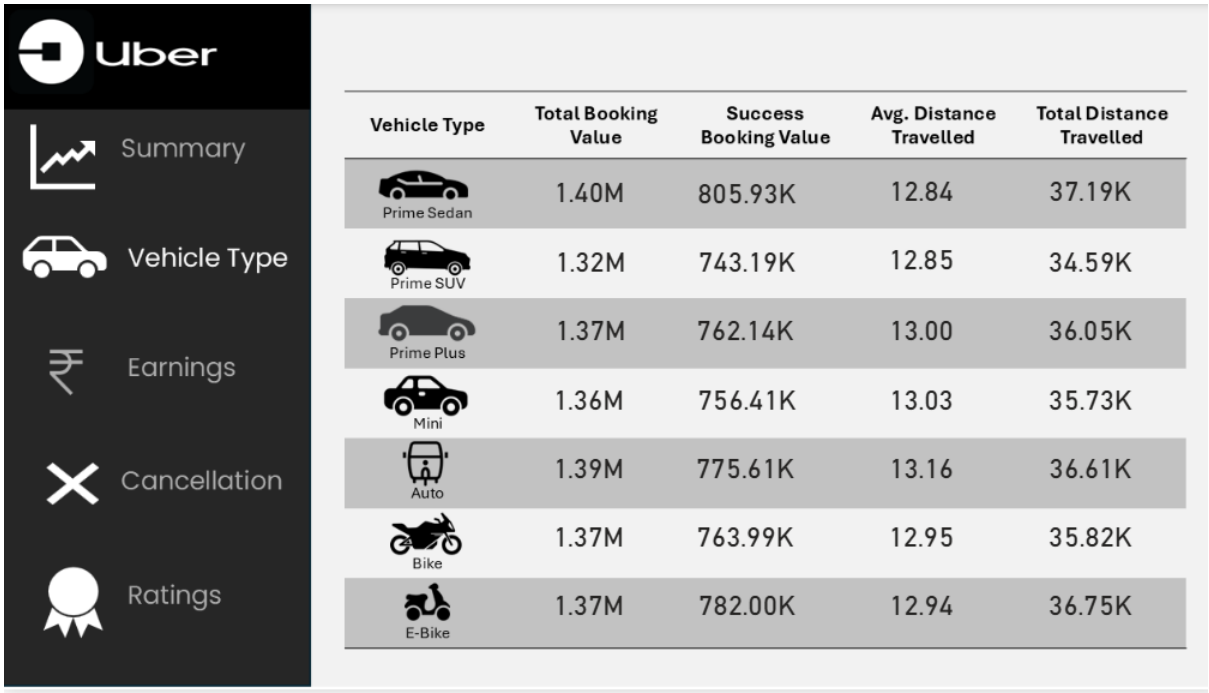
	total_successful_value
▶	5389274.31000003

• Incomplete Rides – Identified all Incomplete Rides along with reason

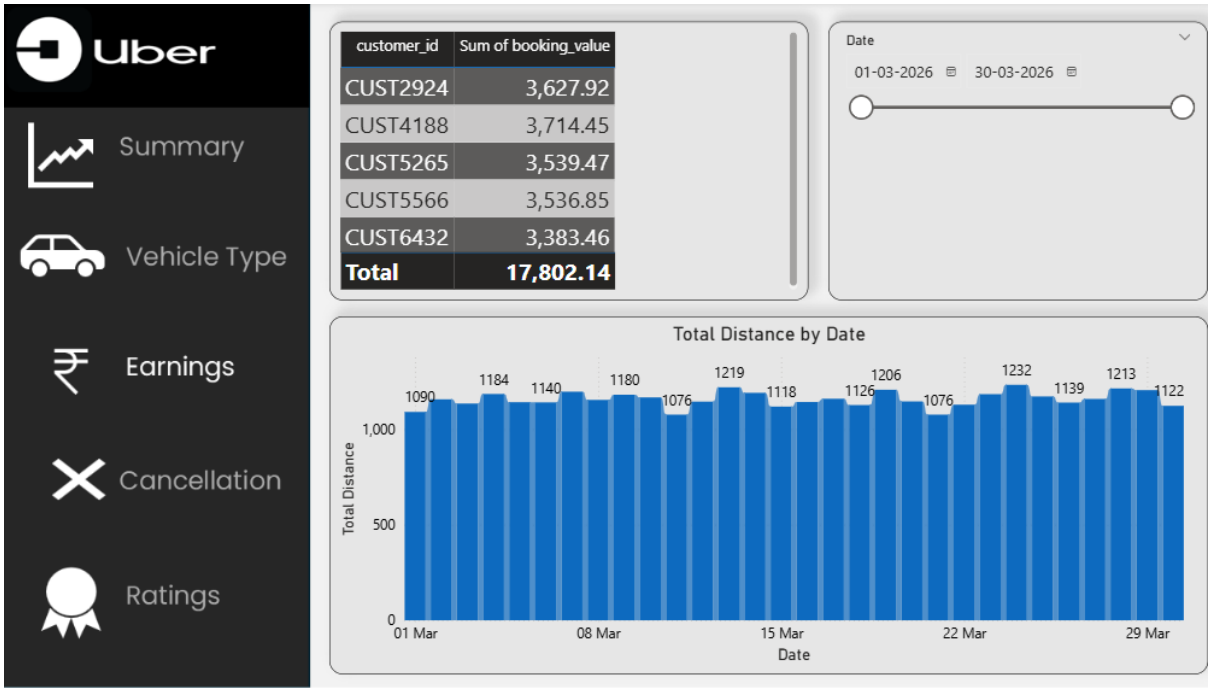
Dashboard in Power BI

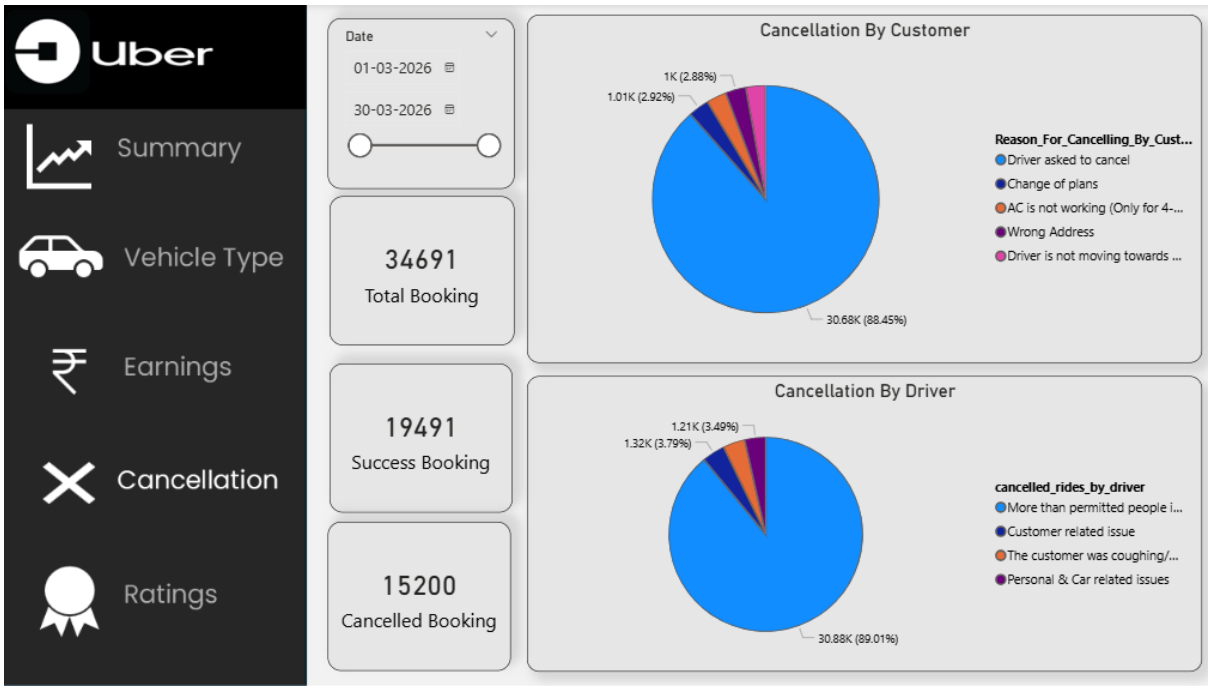
Developed an interactive Power BI dashboard to track key Uber KPIs such as booking success rate, cancellation trends, total revenue, ride distance, and customer & driver ratings. The dashboard enables identification of high cancellation areas, monitoring of vehicle performance, and analysis of customer behavior patterns to support data-driven decisions for improving ride reliability, customer satisfaction, and revenue growth.

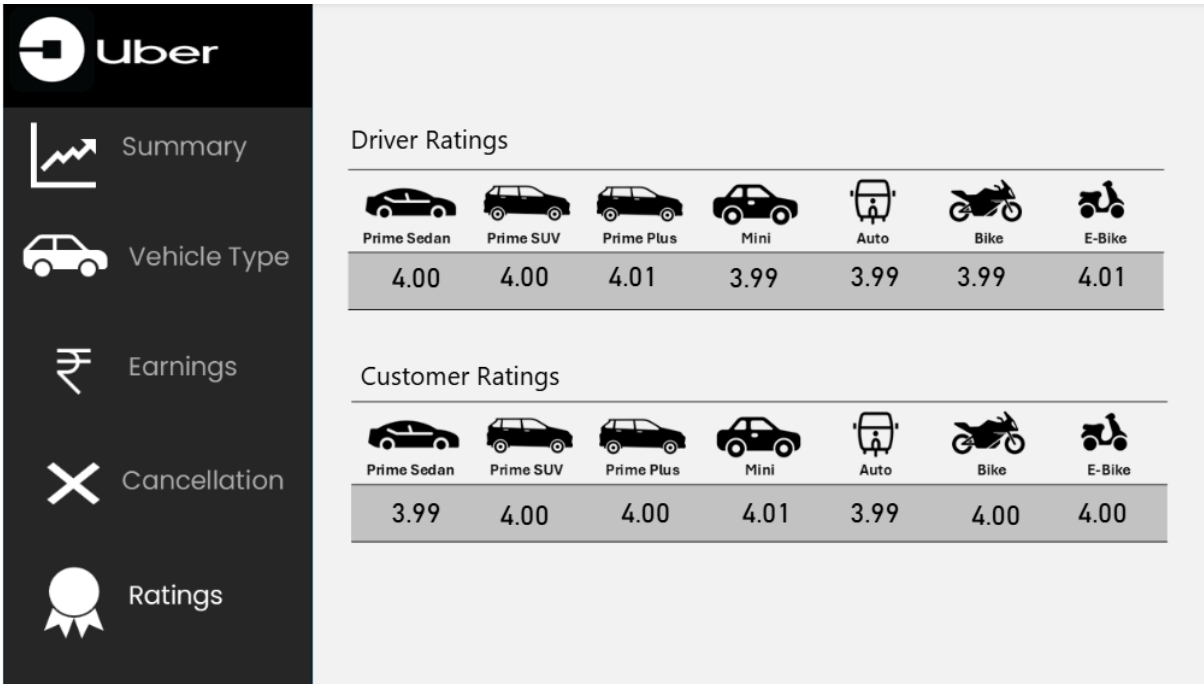




Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	1.40M	805.93K	12.84	37.19K
Prime SUV	1.32M	743.19K	12.85	34.59K
Prime Plus	1.37M	762.14K	13.00	36.05K
Mini	1.36M	756.41K	13.03	35.73K
Auto	1.39M	775.61K	13.16	36.61K
Bike	1.37M	763.99K	12.95	35.82K
E-Bike	1.37M	782.00K	12.94	36.75K


Uber
Summary
Vehicle Type
Earnings
Cancellation
Ratings
customer_id | Sum of booking_value
CUST2924 3,627.92
CUST4188 3,714.45
CUST5265 3,539.47
CUST5566 3,536.85
CUST6432 3,383.46
Total 17,802.14
Date
01-03-2026 30-03-2026
Total Distance by Date
Total Distance
1,000
500
0
1090 1184 1140 1180 1076 1219 1118 1126 1206 1076 1232 1139 1213 1122
01 Mar
08 Mar
15 Mar
22 Mar
29 Mar
Date


Uber
Summary
Vehicle Type
Earnings
Cancellation
Ratings
Date
01-03-2026
30-03-2026
34691
Total Booking
19491
Success Booking
15200
Cancelled Booking
Cancellation By Customer
1K (2.88%)
1.01K (2.92%)
30.68K (88.45%)
Reason_For_Cancelling_By_Cust...
Driver asked to cancel
Change of plans
AC is not working (Only for 4-...
Wrong Address
Driver is not moving towards ...
Cancellation By Driver
1.21K (3.49%)
1.32K (3.79%)
30.88K (89.01%)
cancelled_rides_by_driver
More than permitted people i...
Customer related issue
The customer was coughing/...
Personal & Car related issues



Key Insights from Dashboard:

- Booking Volume: Total number of ride requests (approx. 103K).
- Booking Value: The total gross revenue generated from successful rides (approx. ₹35M).
- Success Rate: The percentage of rides completed successfully (62%).
- Cancellation Rates: Segmented by Customer (27%) and Driver (18%).
- Average Ratings: Split between Customer Rating (3.99) and Driver Rating (3.99), indicating a highly consistent but slightly sub-par service experience.
- Ride Distance: Total distance covered across all successful trips (approx. 1.2M km).

Valuable Business Insights

1. Operational Efficiency & Failures

- The "Success Gap": Only 62% of bookings are successful. With 38% of potential revenue being lost to cancellations and "incomplete" statuses, there is a massive opportunity for optimization.
- Cancellation Drivers: "Customer-related" issues are the leading cause of cancellations, but "Driver-related" issues (like personal/car issues) account for a significant portion of failed trips.

2. Vehicle Performance

- Prime Sedan Dominance: The "Prime Sedan" category often shows the highest volume or value, but the dashboard reveals that "Mini" and "Bike" categories are essential for volume.
- Distance vs. Value: While Bikes cover high volume in short distances, the Sedan categories drive the "Booking Value" KPI, suggesting a need for a balanced fleet strategy.

3. Customer & Driver Behavior

- Rating Parity: Both drivers and customers hover right at the 4.0 mark. Any dip below this usually triggers churn. The business should investigate why the ratings aren't reaching the 4.5+ "Gold Standard" common in the industry.
- Top Customers: A small percentage of customers (Top 5) account for a disproportionate number of bookings. These "Power Users" should be targeted for loyalty programs to protect the most stable revenue stream.

4. Revenue Insights

- Successful Value vs. Total Value: The dashboard emphasizes ₹35M in successful bookings. Comparing this to the "Cancelled Booking Value" would highlight the exact monetary cost of poor ride matching or driver unavailability.

5. Temporal Trends

- Booking Peaks: The line charts show fluctuations in booking volume over time. Identifying these peak "hot zones" allows for dynamic pricing (surge) and better driver positioning to reduce the customer cancellation rate.

Business Recommendation

• Improve ride success rate by reducing cancellations through better driver allocation and faster ride matching.

• Address customer cancellations by reducing wait times and improving ETA accuracy.

• Minimize driver cancellations by introducing incentives and support for vehicle maintenance and availability.

• Focus on improving service quality to increase ratings from 4.0 to 4.5+ through driver training and better customer experience.

• Optimize fleet distribution by balancing high-value vehicles (Prime Sedan) with high-demand options (Mini, Bike).

• Introduce loyalty programs and rewards for frequent users to retain top customers and increase repeat bookings.

• Reduce revenue leakage by identifying and fixing major causes of cancelled and incomplete rides.

• Use peak hour insights to apply dynamic pricing and position more drivers in high-demand areas.

• Monitor and improve incomplete ride issues by fixing operational or technical problems in the booking process.

• Build a real-time monitoring dashboard to track key metrics like success rate, cancellations, and revenue for faster decision-making.